

Representing & Interpreting Data: Card Sort

COMMON CORE STANDARDS:

CCSS.HSS.ID.A.1 Represent data with plots on the real number line (dot plots, histograms, and box plots).

CCSS.HSS.ID.A.3 Interpret differences in shape, center, and spread in the context of the data sets, accounting for possible effects of extreme data points (outliers).

SKILLS USED:

Creating dot plots, histograms, box plots, and interpreting shape of data.

DIRECTIONS:

This resource is used in conjunction with the Representing and Interpreting Data Power Point and "Interpreting the Distribution of Data" Handout. Students will work in small groups, 2 or 3 are my favorite size groups. Match Histograms, Frequency Displays, Box Plots, Shape, and Description of Measures of Center and Spread. The final result will have 8 groups of 5 cards each. Students will work to sort and group the cards.

DIFFERENTIATION:

The differentiation for this activity will be in the heterogeneous grouping with assigned roles. I like to pre-plan my students into color groups and as they walk into the room I hand them a color card. Then I tell them to get into pairs or groups of 3 (your choice!) of which every person has a different color card. If your classes are like mine, you will have to make sure they don't switch color cards to be with their best friends! Announce which roles are for each color card, for ex. Green Cards are the Techies.

Sample Color Coding: Red cards=Advanced Student, Blue Cards=Intermediate Student, Green Cards=Beginner Level Student

Sample Roles:

Recorder- Does all the writing and scratch work- great role for medium student

Techie- Does all the calculating- great role for beginner level student

Leader- Keeps everyone in the group on task and oversees, but does not do all the work – great role for advanced student

Time-Keeper-Reminds group of time every # of minutes

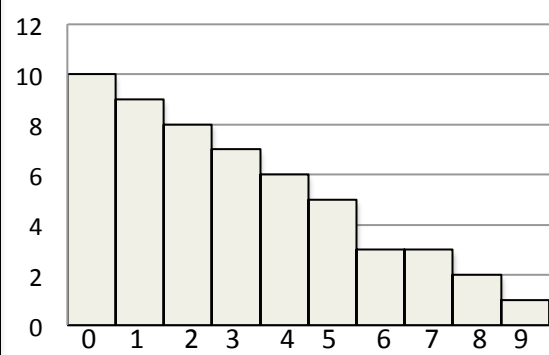
Speaker - I like to ask each group a specific question as a summary and have the middle or beginner level student be the speaker.

Helper-assists each person in the group as well as run errands for the group- another great role for the advanced student

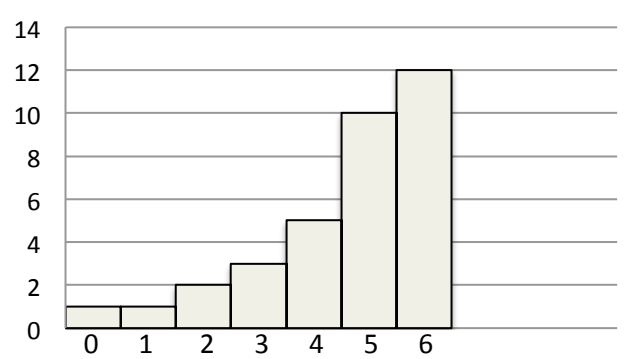
**For this activity, I like the roles of Recorder, Techie, and Helper*

HISTOGRAMS

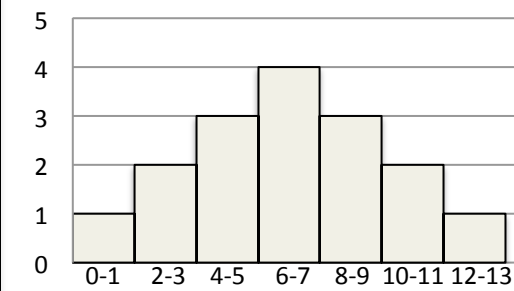
H1



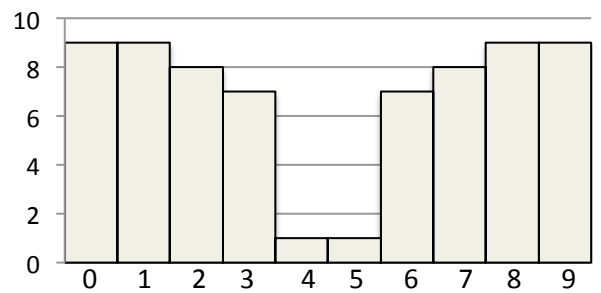
H2



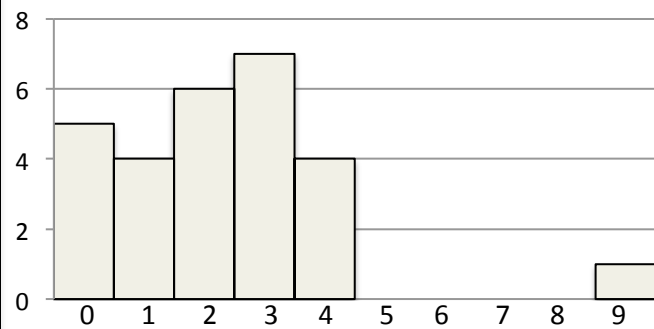
H3



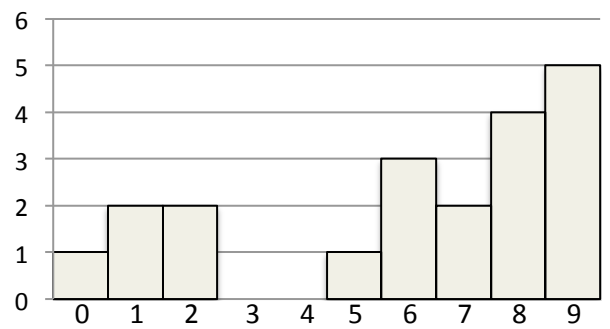
H4



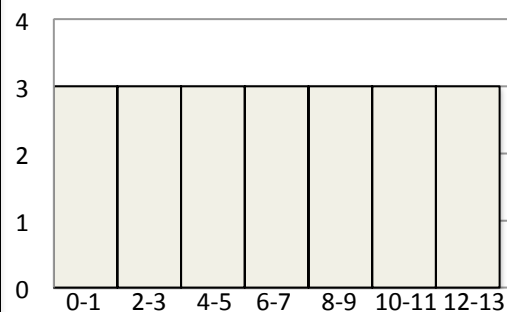
H5



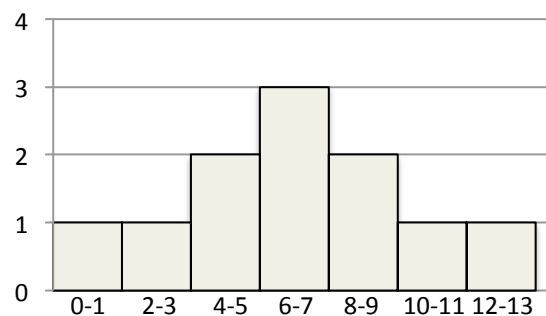
H6



H7



H8

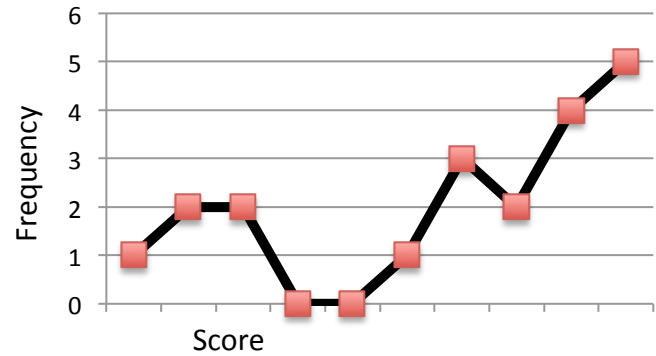


FREQUENCY DISPLAYS

F1

Score	Frequency
0	9
1	9
2	8
3	7
4	1
5	1
6	7
7	8
8	9
9	9

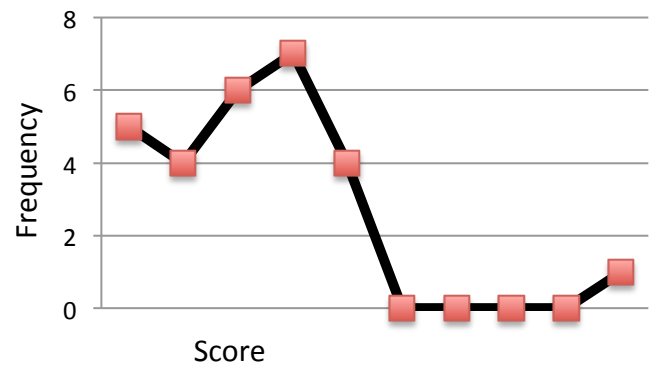
F2



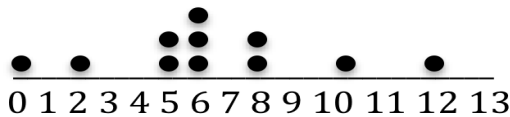
F3

Score	Frequency	Score (contd.)	Frequency (contd.)
0	1	8	2
1	0	9	1
2	1	10	1
3	1	11	1
4	3	12	0
5	0	13	1
6	2		
7	2		

F4



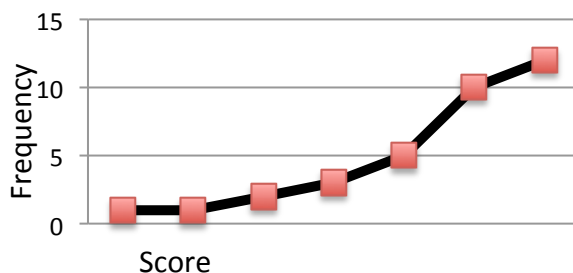
F5



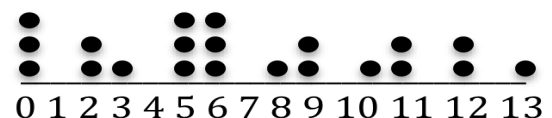
F6

Score	Frequency
0	10
1	9
2	8
3	7
4	6
5	5
6	3
7	3
8	2
9	1

F7

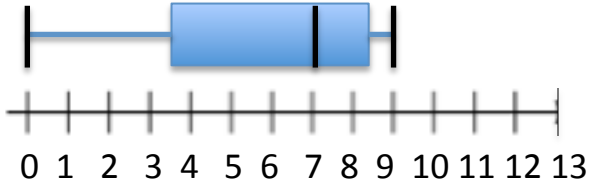


F8

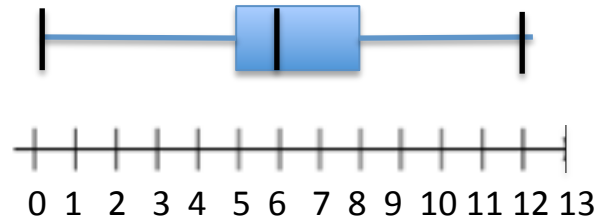


BOX PLOTS

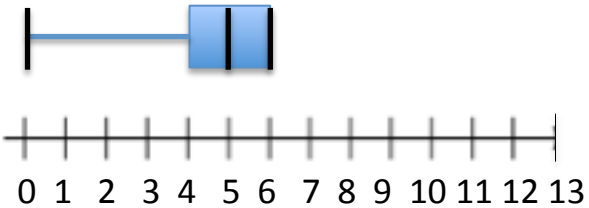
B1



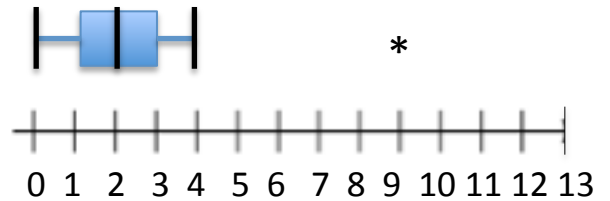
B2



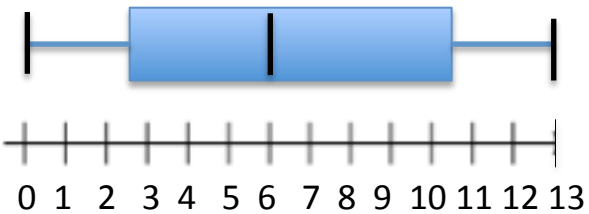
B3



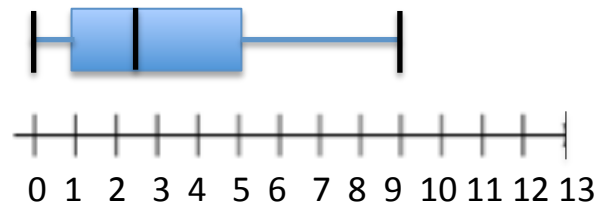
B4



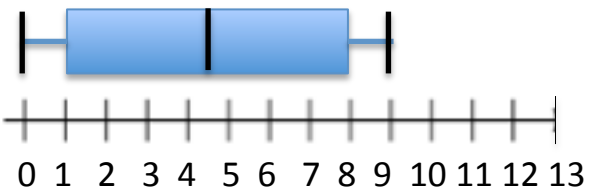
B5



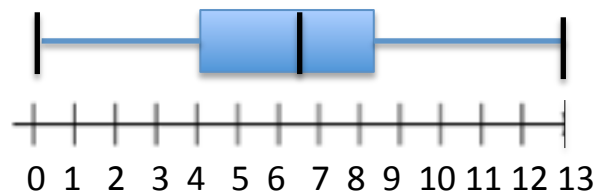
B6



B7



B8



SHAPE – MEASURES OF CENTER/SPREAD

S1 Symmetric U-Shape	S2 Skewed Right
S3 Positive Skew by an Outlier	S4 Asymmetric with a Gap
S5 Histogram Symmetric	S6 Negative Skew
S7 Histogram Perfect Bell Shape	S8 Histogram Uniform
M1 Has 3 Modes	M2 Of the Symmetric histograms, this one has the most variation
M3 IQR=4	M4 Range=6
M5 This data set's mode occurs 5 times.	M6 Range=12
M7 The range is 9 and the IQR is 2.	M8 Measures of Center are Equivalent

Student Recording Sheet

H1	F	B	S	M
H2	F	B	S	M
H3	F	B	S	M
H4	F	B	S	M
H5	F	B	S	M
H6	F	B	S	M
H7	F	B	S	M
H8	F	B	S	M

1. How are dot plots, frequency charts, box plots, and histograms different from each other? Be sure to describe the differences for EACH.
2. What do dot plots, frequency charts, box plots, and histograms all share?
3. Describe why one might prefer to use a histogram to display data.
4. Describe the benefits of using a box plot to display data.
5. What was the hardest part of this activity?
6. After completing the activity, what did you gain a better understanding of?

Answer Key

H1, F6, B6, S2, M3

H2, F7, B3, S6, M4

H3, F3, B8, S7, M2

H4, F1, B7, S1, M8

H5, F4, B4, S3, M7

H6, F2, S4, B1, M5

H7, F8, B5, S8, M1

H8, F5, B2, S5, M6

*Note that some description cards match more than just one group. However, students must find the group for which the description card is the **ONLY** descriptor. This was designed intentionally to generate thinking and discussion amongst student groups 😊

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